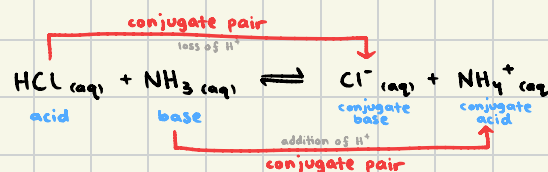


25 - Equilibria

29/12/25

conjugate acid: formed when a base gains a proton

conjugate base: formed when an acid loses a proton



★ pH → related to the hydrogen concentration

* pH is lower for stronger acids & higher for weaker acids

$$\text{pH} = -\log_{10} [\text{H}^+ \text{ (aq)}]$$

hydrogen ion conc.

e.g Calculate the pH of a solution whose H⁺ ion conc. is $5.32 \times 10^{-4} \text{ mol dm}^{-3}$

$$\text{pH} = -\log_{10} [\text{H}^+]$$

$$\text{pH} = -(\log_{10} (5.32 \times 10^{-4}))$$

$$\boxed{\text{pH} = 3.27}$$

Calculate the H⁺ ion conc. of a solution whose pH = 2.90

$$\text{pH} = -\log_{10} [\text{H}^+] \longrightarrow [\text{H}^+] = 10^{-\text{pH}}$$

$$[\text{H}^+] = 10^{-2.9}$$

$$\boxed{[\text{H}^+] = 1.26 \times 10^{-3} \text{ mol dm}^{-3}}$$

★ Ionic product of water (K_w)

equilibrium of water: $\text{H}_2\text{O} \rightleftharpoons \text{H}^+ + \text{OH}^-$

$$\text{equilibrium constant} \Rightarrow K_c = \frac{[\text{H}^+][\text{OH}^-]}{[\text{H}_2\text{O}]}$$

* the extent of ionisation of water is very low thus conc. of H⁺ & OH⁻ is very low & K_c is very low due to which we can regard conc. of water as being constant

as a result we form

$$K_w = [\text{H}^+][\text{OH}^-]$$

! at 298K: pH of distilled water = 7

$$\Rightarrow [\text{H}^+] = 10^{-7}$$

$$[\text{H}^+] = 10^{-7}$$

$$\therefore [\text{OH}^-] = 10^{-7}$$

$$[\text{H}^+] = [\text{OH}^-]$$

$$K_w = (10^{-7})(10^{-7})$$

$$\boxed{K_w = 1.00 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}}$$

at 298K

K_w is a constant for a constant temp

e.g calculate the pH of $0.025 \text{ mol dm}^{-3}$ of NaOH (fully ionises)



$$[\text{NaOH}] = [\text{OH}^-]$$

$$0.025 = [\text{OH}^-]$$

$$K_w = [\text{H}^+][\text{OH}^-]$$

$$1 \times 10^{-14} = [\text{H}^+](0.025)$$

$$[\text{H}^+] = 4 \times 10^{-13}$$

$$\text{pH} = -\log [\text{H}^+]$$

$$\text{pH} = -\log (4 \times 10^{-13})$$

$$\boxed{\text{pH} = 12.4}$$

calculate the pH of $0.025 \text{ mol dm}^{-3}$ of $\text{Ba}(\text{OH})_2$ (fully ionises)



$$2[\text{Ba}(\text{OH})_2] = [\text{OH}^-]$$

$$2(0.025) = [\text{OH}^-]$$

$$K_w = [\text{H}^+][\text{OH}^-]$$

$$1 \times 10^{-14} = [\text{H}^+](0.05)$$

$$[\text{H}^+] = 2 \times 10^{-13}$$

$$\text{pH} = -\log [\text{H}^+]$$

$$\text{pH} = -\log (2 \times 10^{-13})$$

$$\boxed{\text{pH} = 12.7}$$



acid dissociation constant (K_a)

→ shows the extent of dissociation of an acid

→ higher the K_a value, greater the dissociation of acid



$$K_a = \frac{[\text{H}^+][\text{A}^-]}{[\text{HA}]}$$

e.g Calculate the value of K_a for 0.01 mol dm^{-3} of methanoic acid with $\text{pH} = 2.90$



$$[\text{H}^+] = 10^{-\text{pH}}$$

$$[\text{H}^+] = 10^{-2.9} = 1.26 \times 10^{-3} \text{ mol dm}^{-3}$$

$$K_a = \frac{[\text{H}^+][\text{A}^-]}{[\text{HA}]}$$

$$K_a = \frac{(1.26 \times 10^{-3})^2}{0.01} \longrightarrow \boxed{K_a = 1.59 \times 10^{-4} \text{ mol dm}^{-3}}$$

$$\star \text{p}K_a = -\log_{10}(K_a) \star$$

e.g calculate the pH of $0.100 \text{ mol dm}^{-3}$ of ethanoic acid ($K_a = 1.74 \times 10^{-5}$)



$$K_a = \frac{[\text{H}^+]^2}{[\text{CH}_3\text{COOH}]}$$

$$1.74 \times 10^{-5} = \frac{[\text{H}^+]^2}{0.100}$$

$$[\text{H}^+] = \sqrt{(0.1)(1.74 \times 10^{-5})}$$

$$[\text{H}^+] = 1.32 \times 10^{-3} \text{ mol dm}^{-3}$$



$$\text{pH} = -\log_{10} [\text{H}^+]$$

$$\text{pH} = -\log_{10} (1.32 \times 10^{-3}) \longrightarrow \boxed{\text{pH} = 2.88}$$

Buffer solution

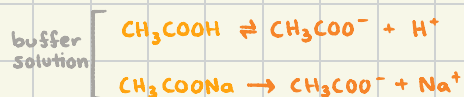
★ a solution that resists changes in pH on addition of small amounts of acid or alkali

★ used to keep pH constant for certain reactions

Types of buffer

Acidic buffer

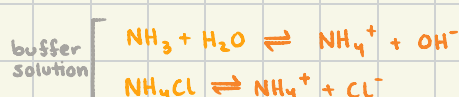
weak acid + salt of its conjugate base



- 1) solution has a relatively high conc. of un-ionised ethanoic acid because its a weak acid so it dissociates only partly
- 2) solution has a relatively high conc. of ethanoate ions which largely are formed from the fully ionised sodium ethanoate & ethanoic acid only gives rise to a very low conc. of ethanoate ions.
- 3) enough H^+ to make the solution acidic

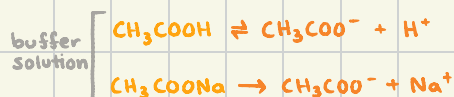
Alkaline buffer

weak base + salt of its conjugate acid

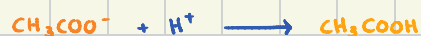


- 1) solution has a relatively high conc. of un-ionised ammonia because its a weak base so it dissociates only partly
- 2) solution has a relatively high conc. of ammonium ions which largely are formed from the fully ionised ammonium chloride & ammonia only gives rise to a very low conc. of ammonium ions.
- 3) enough OH^- to make the solution basic

How does a buffer solution work?



↳ if you add a small amount of acid (H^+)

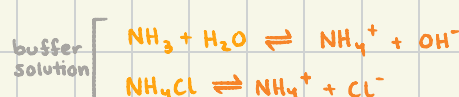


the ethanoate ions present will react with added H^+ to make unionised acid, this is because the addition of H^+ shifted the equilibrium to the L.H.S

↳ if you add a small amount of a base (OH^-)



the H^+ ions present will react with the added OH^- to make water, this reduces the conc. of H^+ , shifting the equilibrium to the R.H.S, causing more CH_3COOH molecules to ionise to form more H^+ (& CH_3COO^-)



↳ if you add a small amount of acid (H^+)



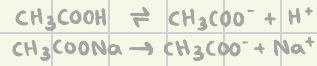
the OH^- ions present will react with the added H^+ to make water, this reduces the conc. of OH^- , shifting the equilibrium to the R.H.S, causing more NH_3 molecules to ionise to form more OH^- (& NH_4^+)

↳ if you add a small amount of a base (OH^-)



the ammonium ions present will react with added OH^- to make unionised base, this is because the addition of OH^- shifted the equilibrium to the L.H.S

e.g. what is the pH of a buffer solution made by 50cm³ of 0.300 mol dm⁻³ of ethanoic acid & 100cm³ of 0.600 mol dm⁻³ of sodium ethanoate [K_a for CH₃COOH = 1.74 × 10⁻⁵]



same as salt

$$K_a = \frac{[\text{CH}_3\text{COO}^-][\text{H}^+]}{[\text{CH}_3\text{COOH}]}$$

same as conc. of unionised acid

$$\text{mol of CH}_3\text{COOH} : \frac{0.3 \times 50}{1000} = 0.0150 \text{ mol (in 150cm}^3\text{)}$$

$$\text{conc. of CH}_3\text{COOH} : \frac{0.0150}{150} \times 1000 = 0.1 \text{ mol dm}^{-3}$$

$$\text{mol of CH}_3\text{COO}^- : \text{equal to CH}_3\text{COONa} \Rightarrow \frac{0.6 \times 100}{1000} = 0.06 \text{ mol (in 150cm}^3\text{)}$$

$$\text{conc. of CH}_3\text{COO}^- : \frac{0.06}{150} \times 1000 = 0.4 \text{ mol dm}^{-3}$$

$$1.74 \times 10^{-5} = \frac{[0.4][\text{H}^+]}{[0.1]} \rightarrow \text{pH} = -\log_{10}(4.35 \times 10^{-6})$$

$$[\text{H}^+] = 4.35 \times 10^{-6} \rightarrow \underline{\underline{\text{pH} = 5.36}}$$

Henderson's equation : $\text{pH} = \text{pK}_a + \log \frac{[\text{salt}]}{[\text{acid}]}$ ★ NOT IN THE SYLLABUS but can be used to solve buffer q's
for buffer solution

e.g. Calculate the pH of a buffer solution containing 0.600 mol dm⁻³ propanoic acid & 0.800 mol dm⁻³ sodium propanoate
[K_a of propanoic acid = 1.35 × 10⁻⁵ mol dm⁻³]



$$K_a = \frac{[\text{C}_2\text{H}_5\text{COO}^-][\text{H}^+]}{[\text{C}_2\text{H}_5\text{COOH}]}$$

$$1.35 \times 10^{-5} = \frac{[0.800][\text{H}^+]}{[0.600]} \rightarrow [\text{H}^+] = \frac{(1.35 \times 10^{-5})(0.600)}{(0.800)}$$

$$[\text{H}^+] = 1.01 \times 10^{-5} \text{ mol dm}^{-3}$$

$$\text{pH} = -\log_{10}(1.01 \times 10^{-5})$$

$$\underline{\underline{\text{pH} = 4.99}}$$

$$[\text{H}^+] = \frac{K_a \times [\text{salt}]}{[\text{acid}]}$$

$$[\text{H}^+] = K_a \times \frac{[\text{salt}]}{[\text{acid}]}$$

$$\text{pH} = \text{pK}_a + \log \frac{[\text{salt}]}{[\text{acid}]}$$

Henderson's equation

↓

$$\text{pH} = -\log(1.35 \times 10^{-5}) + \log \frac{(0.8)}{(0.6)}$$

$$\text{pH} = \log \left[\left(\frac{4}{3} \right) (1.35 \times 10^{-5}) \right]$$

$$\underline{\underline{\text{pH} = 4.99}} \quad \text{Same ans. as}$$

Use of buffer:

↳ used to control pH in many ways.

↳ in our blood, the most important buffer system is the equilibrium b/w CO_2 & HCO_3^- (hydrogen carbonate ion)

Biology recap: CO_2 is a product of aerobic respiration
 H^+ combines with water in the blood (in RBC) (catalysed by carbonic anhydrase)



if H^+ conc. increases

equilibrium shifts to the L.H.S

$\text{H}^+ + \text{HCO}_3^-$ combine to form $\text{CO}_2 + \text{H}_2\text{O}$

conc. of H^+ reduces & equilibrium restores

if H^+ conc. decreases

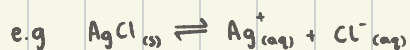
equilibrium shifts to the R.H.S

$\text{CO}_2 + \text{H}_2\text{O}$ combine to form $\text{H}^+ + \text{HCO}_3^-$

conc. of H^+ increases & equilibrium restores

Solubility product (K_{sp}):

↳ a sparingly soluble salt will form an equilibrium with its aqueous ions in water



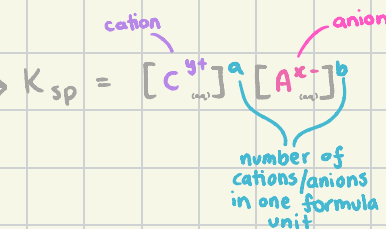
$$K_c = \frac{[\text{Ag}^+][\text{Cl}^-]}{[\text{AgCl}]}$$

the conc. of the solid phase remains constant so it can be combined with K_c → $K_{sp} = [\text{Ag}^+][\text{Cl}^-]$

Calculating unit of K_{sp} :

$$\begin{aligned} K_{sp} &= [\text{Mg}^{2+}][\text{OH}^-]^2 \\ &= (\text{mol dm}^{-3})(\text{mol dm}^{-3})^2 \\ &= \text{mol}^3 \text{dm}^{-9} \end{aligned}$$

$$K_{sp} = [\text{C}^{y+}]^a [\text{A}^{x-}]^b$$

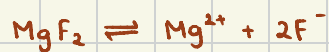


smaller the K_{sp} value, lower solubility of the salt



$$K_{sp} = [\text{Fe}^{3+}]^2 [\text{S}^{2-}]^3$$

e.g. Q: A saturated solution of magnesium fluoride (MgF_2) has a solubility of $1.22 \times 10^{-3} \text{ mol dm}^{-3}$. Calculate the solubility product of magnesium fluoride.



when $1.22 \times 10^{-3} \text{ mol}$ dissolves to form a 1 dm^3 of solution

$$\hookrightarrow [\text{Mg}^{2+}] = 1.22 \times 10^{-3} \text{ mol dm}^{-3}$$

$$\begin{aligned} \hookrightarrow [\text{F}^-] &= 2 \times (1.22 \times 10^{-3}) \text{ mol dm}^{-3} \\ &= 2.44 \times 10^{-3} \text{ mol dm}^{-3} \end{aligned}$$

each formula unit contains $2 \times \text{F}^-$ ions which is why you multiply conc. by two

$$K_{sp} = [\text{Mg}^{2+}][\text{F}^-]^2$$

$$K_{sp} = (1.22 \times 10^{-3})(2.44 \times 10^{-3})^2$$

$$K_{sp} = 7.26 \times 10^{-9}$$

$$\begin{aligned} \text{unit: } & \overset{\text{Mg}}{(\text{mol dm}^{-3})} \times \overset{\text{F}_2}{(\text{mol dm}^{-3})^2} \\ & \rightarrow \text{mol}^3 \text{ dm}^{-9} \end{aligned}$$

$$\underline{\underline{K_{sp} = 7.26 \times 10^{-9} \text{ mol}^3 \text{ dm}^{-9}}}$$

Q: Calculate the solubility of copper (II) sulfide in mol dm^{-3} [K_{sp} for $\text{CuS} = 6.3 \times 10^{-36} \text{ mol}^2 \text{ dm}^{-6}$]



write the equil. expression in terms of one ion only

$$\text{---} \rightarrow [\text{Cu}^{2+}] = [\text{S}^{2-}]$$

$$\therefore K_{sp} = [\text{Cu}^{2+}]^2$$



$$6.3 \times 10^{-36} = [\text{Cu}^{2+}]^2$$

$$[\text{Cu}^{2+}] = \sqrt{6.3 \times 10^{-36}}$$

$$\underline{\underline{[\text{Cu}^{2+}] = 2.5 \times 10^{-18} \text{ mol dm}^{-3}}}$$

in the case of MgF_2

$\hookrightarrow$ if $[\text{Mg}^{2+}] = x$
then $[\text{F}^-] = 2x$

$$\therefore K_{sp} = (x) \cdot (2x)^2$$

$$K_{sp} = 4x^3$$

Common ion effect:

- the reduction in the solubility of a dissolved salt achieved by adding a solution of a compound that has an ion in common with the dissolved
- if a solution containing one of the ions of salt XY is added to a saturated solution of the salt XY

Simply...

the solubility of salt XY is reduced in the presence of ions X or Y from another source



e.g. saturated solution of MgCO_3
 $\text{MgCO}_3 \rightleftharpoons \text{Mg}^{2+} + \text{CO}_3^{2-}$ $\longrightarrow$ if Na_2CO_3 solution is now added, the $[\text{CO}_3^{2-}]$ increases so equilibrium shifts to L.H.S, so MgCO_3 is precipitated out

Partition coefficients

doesn't dissolve in each other

$\rightarrow$ if a particular solute is dissolved in an immiscible mixture of two liquids

$\rightarrow$ equilibrium is established in this case

Ratio of conc. of solute in two immiscible solvents

e.g.



$$K_{pc} = \frac{[\text{I}_2 (\text{hexane})]}{[\text{I}_2 (\text{water})]}$$

(has no units)

e.g. if 0.35g of iodine is shaken with 100cm^3 of water & 100cm^3 of hexane and the mixture is allowed to establish equilibrium, the conc. of iodine in the aq. layer is found to be $4.0 \times 10^{-3} \text{ mol dm}^{-3}$

(i) find the number of moles of iodine used initially

$$n(\text{I}_2) = \frac{\text{mass}}{M_r} = \frac{0.35}{2(126.9)} = 1.38 \times 10^{-3} \text{ mol}$$

(ii) find the number of moles of iodine present in the aqueous layer

$$n = CV = (4.0 \times 10^{-3}) \left(\frac{100}{1000} \right) = 4.0 \times 10^{-4} \text{ mol}$$

(iii) find the concentration of iodine in the hexane layer

$$n(\text{I}_2 (\text{hexane})) = \overset{\text{initial mol of I}_2}{(1.38 \times 10^{-3})} - \overset{\text{mol of I}_2 \text{ in water}}{(4.0 \times 10^{-4})} = 9.79 \times 10^{-4} \text{ mol}$$

$$C = \frac{n}{V} = \frac{9.79 \times 10^{-4}}{0.1} = 9.79 \times 10^{-3} \text{ mol dm}^{-3}$$

(iv) hence find the partition coefficient

$$K_{pc} = \frac{[\text{I}_2 (\text{hexane})]}{[\text{I}_2 (\text{water})]} \longrightarrow K_{pc} = \frac{9.79 \times 10^{-3}}{4.0 \times 10^{-3}} \longrightarrow K_{pc} = 2.4475$$

conc. of $\text{I}_2 (\text{water}) = \frac{4 \times 10^{-4}}{0.1} = 4 \times 10^{-3} \text{ mol dm}^{-3}$

Successive extractions:- using successive extractions with less amount of solvent portions will extract more mass

e.g when 100cm^3 of aq solution containing 0.50g of an organic compound X was shaken with 20cm^3 of hexane, it was found that 0.4g of X was extracted into the hexane

(i) calculate the partition coefficient of X b/w hexane and water

$$K_{pc} = \frac{[X]_{\text{hexane}}}{[X]_{\text{water}}} = \frac{0.4/20}{0.5-0.4/100} = \underline{\underline{20}}$$

(ii) if two 10cm^3 portions of hexane were used instead of a single 20cm^3 portion, calculate the total amount of X extracted & compare this with the amount extracted using the 20cm^3 portion

extraction 1 :
[mass of X
is a]

$$K_{pc} = 20$$

$$20 = \frac{a/10}{\frac{0.5-a}{100}} \rightarrow \frac{20(0.5-a)}{100} = \frac{a}{10}$$

$$200(0.5-a) = 100a$$

$$100 - 200a = 100a$$

$$300a = 100 \Rightarrow \underline{\underline{a = 0.33\text{g}}}$$

extraction 2 :
[mass of X
is b]

$$20 = \frac{b/10}{\frac{0.5-0.33-b}{100}} \rightarrow \frac{20(0.167-b)}{100} = \frac{b}{10}$$

$$2(0.167-b) = b$$

$$0.334 - 2b = b$$

$$3b = 0.334 \rightarrow \underline{\underline{b = 0.111\text{g}}}$$

$$\text{Total extracted} = \underline{\underline{0.444\text{g}}}$$

↓
using 20cm^3
portion extracted
 0.4g

↓
thus, more mass
was obtained by
successive extractions

What affects K_{pc} ?

→ polarity of the solute & solvents

Solute will partition more into the solvent with a 'matching' polarity, increasing its conc. in that layer

→ intermolecular forces (hydrogen bonding, van der Waals forces etc.)

solvent that can form stronger bonds with the solute will hold more of it